DTC P1051:00 [SKYACTIV-D 1.5]

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Note
- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.

Details On DTCs

DESCRIPTION	Injection quantity too low cylinder No.1	
DETECTION CONDITION	Determination conditions	• Small torque fluctuation condition continues in No.1 cylinder while main injection is performed 40 times in cylinder No.1 by fuel injector No.1
	Preconditions	• Vehicle speed: **2 km/h {2 mph}** or more • APP: **0.1% or less** • Engine speed: **850rpm or less** • ECT: **40 °C {104 °F} or more** • IAT (IAT sensor No.1): **-10 °C {14 °F} or more** • Fuel pressure at common rail: **35 MPa {357 kgf/cm^2, 5077 psi} or more** • During fuel injection • Diesel particulate filter regeneration control is not performed • Other than moment clutch is engaged or released (MTX) • Other than moment gear is shifted (ATX) • The following DTC is not detected: — Fuel injector or related circuit malfunction: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00 — Internal PCM malfunction: P062B:00 — ECT sensor or related circuit malfunction: P0116:00, P0117:00, P0118:00 — IAT sensor or related circuit malfunction: P0111:00, P0112:00, P0113:00 — BARO sensor or related circuit malfunction: P2227:00, P2228:00, P2229:00 — CKP sensor or related circuit malfunction: P0336:00, P0337:00
	Drive cycle	• **2**
	Self test type	• CMDTC self test
	Sensor used	• CKP sensor • ABS wheel speed sensor • DSC HU/CM • APP sensor • ECT sensor • IAT sensor No.1 • Fuel pressure sensor
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Check engine light is illuminated	
POSSIBLE CAUSE	• Erratic signal to PCM — ECT sensor signal malfunction — IAT sensor No.1 signal malfunction — CKP sensor signal malfunction — APP sensor signal malfunction — VSS signal malfunction — Fuel pressure sensor malfunction — Input signal part connector or terminals malfunction — Input signal part related circuit malfunction • Fuel shortage in fuel tank • CKP sensor malfunction • Fuel delivery system malfunction — Fuel restriction in fuel line • Fuel injector No.1 malfunction • Mechanical (engine) malfunction — Insufficient engine compression — Engine coolant leakage to combustion chamber • Poor fuel quality • PCM malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The PCM performs diagnosis of the fuel injector amounts by comparing the engine torque fluctuation occurring at the engine stroke between each cylinder.
- The PCM calculates the engine torque fluctuation by the change in the angular speed of the crankshaft from TDC to BDC of the expansion stroke of each cylinder.
- When the monitor conditions are met, the PCM compares the engine torque fluctuation between each cylinder, and if the No. 1 cylinder torque fluctuation (crankshaft angular speed) is less than the specified value, it determines that the fuel injection amount of fuel injector No. 1 has decreased.
- The PCM performs diagnosis when each of the preconditions during drive cycle is met. If a malfunction is detected, the malfunction detection counter begins counting up. The value of the malfunction detection counter is maintained while the preconditions are not being met. If the cumulative value of the malfunction detection counter exceeds the threshold during the first drive cycle, a malfunction is determined and a pending code is stored.
- If the PCM determines that the malfunction recurs from the next drive cycle and thereafter, it stores a DTC and turns on the check engine light.

Repeatability Verification Procedure

Caution
- **When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.**
- **While performing this step, always operate the vehicle in a safe and lawful manner.**

1. Start the engine and warm it up until the engine coolant temperature is 40 °C {104 °F} or more.
2. Accelerate to 40 km/h {25 mph}.
3. Release the clutch while staying in gear and decelerate (MTX). Decelerate without shifting from the D position to another position (ATX).
4. Continue to maintain the engine speed at 850 rpm or less for a cumulative 10 s or more at a speed of 2 km/h {2 mph} or more during deceleration.

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
APP1	Accelerator pedal position No.1 voltage	V	Switch ignition ON (engine off) • Accelerator pedal released: Approx. 0.79 V • Accelerator pedal depressed: Approx. 4.54 V
APP2	Accelerator pedal position No.2 voltage	V	Switch ignition ON (engine off) • Accelerator pedal released: Approx. 0.38 V • Accelerator pedal depressed: Approx. 2.27 V
ECT	Engine coolant temperature	°C, °F	Displays ECT
IAT	Intake air temperature No.1	°C, °F	Displays the intake air temperature (No.1)
FRP	Fuel pressure at common rail	KPa {MPa}, mBar {Bar}, psi, in H20	• Idle: Approx. 40 MPa {408 kgf/cm^2, 5802 psi} • Racing (engine speed above 4,000 rpm): 90 MPa {918 kgf/cm^2, 13053 psi}
VSS	Vehicle speed	KPH, MPH	This PID indicates vehicle speed.
WSPD_SEN_LF	LF ABS wheel speed sensor	KPH, MPH	This PID indicates vehicle speed.
WSPD_SEN_LR	LR ABS wheel speed sensor	KPH, MPH	This PID indicates vehicle speed.
WSPD_SEN_RF	RF ABS wheel speed sensor	KPH, MPH	This PID indicates vehicle speed.
WSPD_SEN_RR	RR ABS wheel speed sensor	KPH, MPH	This PID indicates vehicle speed.

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P1051:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
3	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY OTHER RELATED DTCs OCCURRING** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the other PENDING CODE/DTC also present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**PURPOSE: VERIFY IF THERE IS PID ITEM CAUSING DRASTIC CHANGES OF ACCELERATION FLUCTUATION BY INPUT SIGNAL TO PCM AND DSC HU/CM** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See PID/DATA MONITOR INSPECTION [DSC HU/CM].) **PCM:** — APP1 — APP2 — IAT — ECT **DSC:** — WSPD_SEN_LF — WSPD_SEN_LR — WSPD_SEN_RF — WSPD_SEN_RR • Is there any signal that is far out of specification?	Yes	Inspect the suspected sensor and related wiring harness. • If there is any malfunction: — Repair or replace the malfunctioning part. — Go to the troubleshooting procedure to perform the procedure from Step 6. • If there is no malfunction: — Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**PURPOSE: VERIFY CONNECTOR CONNECTIONS** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See PID/DATA MONITOR INSPECTION [DSC HU/CM].) **PCM:** — APP1 — APP2 — IAT — ECT **DSC:** — WSPD_SEN_LF — WSPD_SEN_LR — WSPD_SEN_RF — WSPD_SEN_RR • When the following parts are shaken, does the PID value include a PID item which has changed? — APP sensor — IAT sensor No.1 — ECT sensor — LF ABS wheel speed sensor — LR ABS wheel speed sensor — RF ABS wheel speed sensor — RR ABS wheel speed sensor — PCM — DSC HU/CM	Yes	Repair or replace the applicable connector parts. Go to the troubleshooting procedure to perform the procedure from Step 6.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Perform inspection of lack of fuel.
- Step 2—4
  - — Perform inspection of each separate part.
- Step 5
  - — Perform inspection of engine.
- Step 6—7
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION		ACTION
1	**PURPOSE: DETERMINE IF MALFUNCTION IS DUE TO LACK OF FUEL** • Verify the remaining amount of fuel in the fuel tank. • Is there enough fuel in the fuel tank?	Yes	Go to the next step.
		No	Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **If DTC P1051:00 is displayed:** • Go to the next step. **If DTC P1051:00 is not displayed:** • Troubleshooting is completed. (The cause of this DTC is low fuel level)
2	**PURPOSE: INSPECT CKP SENSOR** • Inspect the CKP sensor. (See CRANKSHAFT POSITION (CKP) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the CKP sensor, then go to Step 6. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**PURPOSE: INSPECT FOR FUEL LEAKAGE FROM FUEL SYSTEM** • Visually inspect the followings: — Fuel leakage from the fuel tank, fuel pump, hose, pipe, fuel injector, supply pump, common rail — Cracking and damage in fuel hose and pipe — Clamp installation condition for each hose and pipe — Fuel pipe securing condition due to deterioration such as rubber of clamp • Are all items normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results, then go to Step 6.
4	**PURPOSE: INSPECT FUEL INJECTOR NO.1** • Inspect the fuel injector No.1. (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the fuel injector No.1, then go to Step 6. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**PURPOSE: INSPECT FOR MALFUNCTION DUE TO ENGINE** • Inspect the engine for followings: — Engine compression (See COMPRESSION INSPECTION [SKYACTIV-D 1.5].) — Engine coolant leakage (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to the next step.
		No	A malfunction caused by the use of inferior fuel can be considered. Replace the fuel, then go to the next step. (See FUEL DRAINING PROCEDURE [SKYACTIV-D 1.5].)
6	**PURPOSE: PERFORM DTC INSPECTION AND VERIFY IF MALFUNCTIONING PART IS PCM** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
7	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.